

MANASWINI GOGINENI

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PROFESSIONAL SUMMARY

Innovative Software Developer with 3+ years of experience building scalable cloud-native applications, microservices, and serverless systems using Python, Go, Java, C++, JavaScript, and TypeScript. Skilled in full-stack development, APIs, containerization, CI/CD pipelines, AWS, Kubernetes, Docker, Terraform, databases, observability, and automated testing.

TECHNICAL SKILLS

Programming Languages: Python, Go, Java, C++, C, JavaScript (ES6+), TypeScript, SQL, Bash
Frameworks & Libraries: React, Next.js, Node.js, Express.js, Spring Boot, Angular, Vue.js, React Native, Tailwind CSS, Redux Toolkit, FastAPI, Flask
Cloud & DevOps: AWS (Lambda, ECS, EKS, S3, CDK, CloudFormation, CloudWatch), Azure, GCP, Docker, Kubernetes, Terraform, Ansible, Jenkins, GitHub Actions, ArgoCD, Helm, Datadog, Prometheus, Grafana, Gremlin
Databases & Storage: PostgreSQL, MySQL, DynamoDB, MongoDB, Redis, SQLite, Cassandra, S3, Elasticsearch
Automation, QA & Testing: Cypress, Playwright, Selenium, PyTest, JUnit, Postman, Newman, SonarQube, Jest, Mocha, Chai
Systems & Embedded Development: C/C++, Linux Systems Programming, Multithreading, Socket Programming, RTOS Concepts,
APIs & Integration: RESTful APIs, GraphQL, gRPC, WebSockets, Kafka, RabbitMQ, OpenAPI/Swagger
AI & Data Tools: Pandas, NumPy, TensorFlow Lite, OpenAI API, LangChain, Hugging Face Transformers (for AI-augmented applications)
Version Control & Collaboration: Git, GitHub, GitLab, Bitbucket, Jira, Confluence, Agile/Scrum
Build & CI/CD Tools: Maven, Gradle, NPM, Yarn, Webpack, Vite, CircleCI
Security & Monitoring: IAM, Vault, CloudTrail, OWASP, Snyk, Static Code Analysis, SIEM Integration

PROFESSIONAL EXPERIENCE

Software Developer Intern Jun 2025 - Aug 2025
Cisco ThousandEyes

- Developed a scalable integration dashboard using React, TypeScript, and Tailwind CSS to visualize multi-source telemetry, enhancing team efficiency and reducing analysis time by 30%.
- Built and optimized serverless backend services and GraphQL APIs with Node.js, Express, and AWS Lambda, improving data ingestion speed by 40% and reducing manual workflows by 70%.
- Automated webhook processing and CI/CD pipelines with Docker, Kubernetes, and GitHub Actions, achieving 99.9% uptime and reducing synchronization errors

Python Developer Co-op Jun 2024 - Dec 2024
Electric Hydrogen

- Developed a Python-based telemetry and monitoring system using AWS Lambda, CDK, and Slack Webhooks, improving real-time visibility and reducing cloud monitoring costs by 30%.
- Enhanced ETL reliability by 40% through DLQ-enabled Lambda → S3 ingestion workflows and automated alerting mechanisms for proactive data management.
- Optimized database performance and analytics speed by 35% via Python automation, PostgreSQL schema tuning, and modular microservice design for scalable backend operations.
- Streamlined CI/CD and infrastructure provisioning using Terraform, Boto3, and GitHub Actions, cutting deployment lead times by 25% and ensuring consistent, reproducible environments.

Software Engineer II Aug 2021 - Aug 2023
JPMorgan Chase & Co

- Engineered cloud migration of legacy financial services to AWS using Terraform and CDK, reducing provisioning time by 20% and ensuring consistent multi-environment deployments.
- Boosted service reliability by 35% through Lambda/ECS optimization with ALB health checks, Step Functions, and DLQ-based recovery for high-load transaction systems.
- Developed CI/CD pipelines with GitHub Actions and SonarQube, cutting failed deployments by 30% while improving code quality and release stability.
- Refactored monolithic applications into Node.js microservices and containerized them via Docker and ECR, enhancing scalability and accelerating feature delivery

Software Engineer Intern Mar 2021 - Jul 2021
JPMorgan Chase & Co

- Developed a financial reporting platform using GoLang and AngularJS, automating onboarding analysis workflows and reducing manual reporting effort by 30%.
- Automated containerized deployments on AWS EKS with Docker and Jenkins CI/CD pipelines, accelerating release cycles by 40% and improving environment consistency.
- Enhanced system reliability by building Go-based RESTful APIs and implementing unit/integration testing with Jest and Mocha, improving overall application stability.

PROJECTS

Serverless Data Processing Pipeline

- Built a highly available ETL pipeline using AWS Lambda, S3, and DynamoDB, processing streaming data from multiple sources with automated scalability and ensuring 99.9% data reliability.
- Automated infrastructure deployment with Terraform and AWS CDK and designed CloudWatch dashboards to monitor metrics and detect anomalies in real-time, reducing manual intervention.

Distributed Cache with Consistent Hashing

- Developed a fault-tolerant distributed key-value store in Go with consistent hashing and replication, achieving sub-10ms read/write latency under benchmark tests.
- Implemented write-ahead logging and LRU eviction policies, containerized nodes with Docker, and orchestrated clusters using Kubernetes, optimizing memory usage and ensuring high durability.

EDUCATION

M.S.E. in Computer Science Aug 2023 - Dec 2025
University of Wisconsin | Madison

CERTIFICATIONS

- AWS Certified Solutions Architect - **Associate**
- CKAD (Certified Kubernetes Application Developer) Specialization - **Linux foundation**
- Generative AI for Software Development
- Introduction to Containers with Docker, Kubernetes & OpenShift - **Coursera**
- Architecting Solutions on AWS - **Coursera**

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Hiring Committee
College of Engineering – Robotics Department
University of Michigan
Ann Arbor, MI

Dear Hiring Committee,

I am excited to apply for the Applications Programmer Associate position with the College of Engineering's Robotics Department at the University of Michigan. With a strong foundation in software development, hands-on experience supporting production applications, and a passion for building tools that empower users, I am eager to contribute to Michigan Engineering's mission of delivering innovative, people-first technology solutions for education and research.

I am currently completing my M.S. in Computer Science at the University of Wisconsin–Madison (Dec 2025), and I have over three years of experience developing full-stack and backend applications in Python, JavaScript/TypeScript, and Go. Across my internships and professional roles, I have built scalable applications, improved system reliability, and supported users by translating their needs into intuitive technology solutions.

During my recent internship at Cisco ThousandEyes, I built internal dashboards and automation tools using React, TypeScript, Node.js, and AWS, which reduced manual workflows by 70%. I collaborated closely with engineering and support teams to gather requirements, iterate on features, and ensure the tools aligned with user expectations—experiences that closely mirror this role's focus on requirement gathering, coding, and team collaboration.

Previously at Electric Hydrogen, I developed Python-based automation, improved data workflows by 40%, and enhanced monitoring systems for better visibility and issue resolution. I frequently handled troubleshooting, root-cause analysis, and documentation—strengthening my ability to provide responsive and clear application support. These experiences have equipped me with the problem-solving skills and communication needed to work effectively with non-technical faculty, staff, and academic users.

I am particularly drawn to this role because of its educational mission and impact. I value the opportunity to support tools that directly enhance the Robotics department's teaching, learning, and research activities. The blend of application development, support, security, and continuous improvement fully aligns with the work I enjoy most—building maintainable software with meaningful user outcomes.

Key strengths I bring to this role include:

- ✓ Proficiency in Python and JavaScript, with experience building custom web applications and internal tools
- ✓ Strong problem-solving and debugging skills supported by hands-on application support experience
- ✓ Ability to collaborate with diverse stakeholders, translate user needs into features, and deliver reliable solutions
- ✓ Self-direction, organization, and ownership of tasks in dynamic, fast-paced environments

I would welcome the opportunity to contribute to the Robotics department's software ecosystem and to support the University of Michigan's commitment to innovation, inclusion, and service. Thank you for considering my application. I look forward to the possibility of discussing how my Technical skillset and problem-solving mindset would make me a strong fit for your team.

Sincerely,
Manaswini Gogineni